UNIT 2: SHAPES AND REGIONS

SESSION 8: BOUNDARY DESCRIPTORS

SLO 1: EXPLAIN DIFFERENT TYPES OF BOUNDARY DESCRIPTORS USED FOR SHAPE REPRESENTATION IN BINARY IMAGES.

Activity 1 – Find the Starting Pixel

Observe the binary image below.

0 0 0 0 0

0 1 1 1 0

0 1 0 1 0

0 1 1 1 0

0 0 0 0 0

Task

1. **Identify the first boundary pixel (top-most, left-most).**

Starting coordinate: (2,2)

Activity 2 – Chain Code Direction Map

Use the 8-direction chain code system.

Direction	Code
Right	0
Down-Right	1
Down	2
Down-Left	3
Left	4
Up-Left	5
Up	6
Up-Right	7

Activity 3 – Trace the Boundary □

Starting from the identified pixel, move clockwise along the boundary.

Write the first six chain codes representing the movement.

Step 1 → 0

Step 2 → 0

Step 3 → 2

Step 4 → 2

Step 5 → 4

Step 6 → 4